

Ousmane THIONGANE

AI & Robotics Engineer looking for a full-time position. Interested in computer vision, robotic perception, and reinforcement learning for control.

📍 France ✉️ ousmane.thiongane@outlook.com ☎️ +33 (0)6 78 18 48 00 💻 ousmane-thiongane 📦 Mowibox

EDUCATION

Three-year Postgraduate Engineering Degree in Electronics and Computer Science

Cergy, France

ENSEA (Ecole Nationale Supérieure de l'Électronique et de ses Applications)

Sep 2022 – Dec 2025

- Major in Computer Science, Signal Processing and Electronics. Specialization in Image Processing & Virtual Reality.
- Head of the robotics association – Competed in the French Robotics Cup and school hackathons; organized hands-on workshops on embedded programming, electronic PCB routing, and soldering.

Exchange Program in Artificial Intelligence & Robotics

Rome, Italy

Sapienza University of Rome

Sep 2024 – Feb 2025

- Coursework:** Artificial Intelligence, Robotics, Computer Vision, Reinforcement Learning and Machine Learning.
- Winner of the “Why did you choose Sapienza?” competition – Public demonstration of a self-designed quadruped robot.

SKILLS

Programming Languages: Python (PyTorch, Keras, Pinocchio, Scikit-learn, Gym, OpenCV, Pytest), C/C++, HTML/CSS.

Other Skills: Image Processing, Deep Learning, Control, Optimization, Camera Calibration, Foundations Models, ROS2, Docker, STM32 Cube IDE, Linux, Git/GitHub, CI/CD, PCB Design (KiCAD), VSCode, Unity 3D, Linear Algebra, Autodesk Sketchbook.

EXPERIENCE

Robotics R&D Engineer

Bordeaux, France

Hugging Face x Pollen Robotics

Mar 2025 – Sep 2025

- Formulation and deployment of a constrained inverse kinematics solver via quadratic programming for the teleoperation of the Reachy2 humanoid robot. Implemented a 500 Hz control loop in Python, with validation in ROS2 simulation and deployment to physical hardware.

Systems Engineering Assistant/Physical Anti-Intrusion Solutions

Osny, France

IDEMIA

Apr 2024 – Aug 2024

- Designed a Python software pipeline for sensor signal filtering and time-series classification (CNN) to detect physical intrusions against defined threat models. Achieved a 97% F1-score with millisecond-level inference latency.

Product Testing & Qualification Engineering Intern

Osny, France

IDEMIA

Jun 2023 – Aug 2023

- Conducted functional qualification protocols on biometric terminals and drafted test reports based on test plans. Developed an image processing pipeline using OpenCV for the extraction and analysis of biometric fingerprints.

PROJECTS

Checkers-RL | *Reinforcement Learning*

<https://github.com/Mowibox/Checkers-RL>

Developed a Reinforcement Learning agent in Python based on $TD(\lambda)$ Linear Value Function Approximation (LVFA). Evaluated the policy against a Monte-Carlo Tree Search (MCTS) baseline. Reduced the policy's relative loss rate by 48% via reward shaping.

ArUco Map Locator | *Computer Vision*

<https://github.com/Mowibox/aruco-map-locator>

Development of a ROS2 visual odometry node. Implemented ArUco fiducial marker detection and algorithmically validated 6DoF pose estimation via Unity3D simulation. Achieved sub-millimeter precision at a stable 30 FPS.

Chromapi | *Robotics*

https://github.com/Mowibox/chromapi_motherboard

End-to-end design of a quadruped robot utilizing a hybrid Raspberry Pi and STM32 architecture. Engineered a custom motherboard for low-level control, interfaced with a software stack to experiment with robotics perception and locomotion models.

LANGUAGES

French – Native

English – Proficient, C1 Level | (TOEIC 910/990)

Spanish – Working Knowledge, B1 Level

References available on request.